

This is unreleased documentation for Serverpod **Next** version.

For up-to-date documentation, see the [latest version](#) (4.0.0).

3.6.5

UPGRADE TO PGVECTOR

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Upgrade to pgvector

New Serverpod projects automatically included pgvector support through the `pgvector/pgvector` PostgreSQL Docker image. From 4.0.0-beta.2, the compose file uses `ghcr.io/serverpod/postgres:16` instead, which is that image plus PostGIS. Existing projects need to be upgraded to use vector functionality.

INFO

This upgrade is only necessary if you want to use vector fields in your models. If you do not plan to use vector fields, you can skip this upgrade.

WARNING

If trying to use vector fields without upgrading, you will encounter an error when applying migrations.

For Docker-based environments

1. Update your `docker-compose.yml` to use a PostgreSQL image with pgvector (e.g., `pgvector/pgvector:pg16`):

```
services:
  postgres:
    image: pgvector/pgvector:pg16 # ← Change from postgres image here
    ports:
      - '8090:5432'
    environment:
      POSTGRES_USER: postgres
      POSTGRES_DB: <projectname>
      POSTGRES_PASSWORD: <DB_PASSWORD>
    volumes:
      - <projectname>_data:/var/lib/postgresql/data

# Other services...
```



```
postgres_test:
  image: pgvector/pgvector:pg16 # ← Change from postgres image here
  ports:
    - '9090:5432'
  environment:
    POSTGRES_USER: postgres
    POSTGRES_DB: <projectname>_test
    POSTGRES_PASSWORD: <DB_TEST_PASSWORD>
  volumes:
    - <projectname>_test_data:/var/lib/postgresql/data
```

2. Recreate your containers to use the new image:

```
docker compose down
docker compose up -d
```

3. Create your first vector field in a model:

```
class: Document
table: document
fields:
  content: String
  embedding: Vector(1536)
```

4. Generate and apply a migration:

```
$ serverpod create-migration
$ dart run bin/main.dart --apply-migrations
```

For more details on creating and applying migrations, see the [Migrations](#) page.

The pgvector extension will be automatically enabled during the first migration that includes a vector column.

For managed PostgreSQL services

For cloud providers (AWS RDS, Google Cloud SQL, Azure Database, etc.), ensure that **pgvector extension is available** on your PostgreSQL instance. Most managed services already support pgvector with no additional setup required. If available, the extension will be enabled automatically when applying the migration.

TIP

If the cloud provider instructions is to run a `CREATE EXTENSION vector;` command, you can skip this step as Serverpod will handle it automatically during migration.

Troubleshooting

If you encounter issues with pgvector:

- Verify that your PostgreSQL version supports pgvector 0.7.0+ (PostgreSQL 12+).
- Check that the pgvector extension is properly installed.
- Ensure your database user has the necessary permissions to create extensions.